

Map_Function

Map

Python's `map()` is a built-in function that enables the processing and transformation of all items in an iterable without the need for an explicit for loop, a technique referred to as mapping. This function is particularly useful when you want to apply a transformation function to each element in an iterable, producing a new iterable as a result. `map()` is one of the tools that facilitate a functional programming approach in Python.

Syntax:-

```
iterable1
iterable2
iterable3

def func_name(n1,n2,n3):
    fun_body...
    fun_body...
    fun_body...
    return final achive value

res = map(func_name,iterable1,iterable2,iterable3)
print(list(res))
```

Examples:

```
my_list=[10,20,30,40]
```

```
def sqr(n):
    return n*n
```

```
x=map(sqr,my_list)
print(x)
print(list(x))
```

O/P:--

```
<map object at 0x000001EA310E3490>
[100, 400, 900, 1600]
```

```
my_tuple=(10,20,30,40)
```

```
def sqr(n):
    return n*n
```

```
x=map(sqr,my_tuple)
print(x)
print(tuple(x))
```

O/P:-

```
<map object at 0x0000019833A83490>
(100, 400, 900, 1600)
```

```
my_str="Neeraj"
def add(n):
    x=ord(n)
    return x
x=map(add,my_str)
print(x)
print(list(x))
```

O/P:-

```
<map object at 0x000001D03A4E3490>
[78, 101, 101, 114, 97, 106]
```

```
my_str="Neeraj"
def add(n):
    x=ord(n)
    return chr(x+5)
x=map(add,my_str)
print(x)
print(list(x))
```

O/P:-

```
<map object at 0x0000026D8F1634C0>
['S', 'j', 'j', 'w', 'f', 'o']
```

```
def add_three(a, b, c):
    return a + b + c
```

```
list1 = [1, 2, 3]
list2 = [4, 5, 6]
list3 = [7, 8, 9]
```

```
result = map(add_three, list1, list2, list3)

print(list(result))
```

O/P:-

[12, 15, 18]

```
def add_three(a, b, c):  
    return a + b + c
```

```
list1 = [1, 2]
```

```
list2 = [4, 5, 6, 7]
```

```
list3 = [7, 8, 9]
```

```
result = map(add_three, list1, list2, list3)
```

```
print(list(result))
```

O/P:-

[12, 15]